

SymPy Bug Card

Bug 4: real antiderivative gains an imaginary term

Task. Compute an indefinite integral and check that it differentiates back:

$$\int \frac{\log x}{x^2 - 1} dx.$$

SymPy's answer. *SymPy* returns an expression F for which

$$F'(2) = \frac{\log 2}{3} - \frac{\pi i}{2}.$$

Correct answer. A valid local antiderivative must satisfy

$$F'(2) = \frac{\log 2}{3}$$

Explanation. The integrand $\log(x)/(x^2 - 1)$ is real and analytic near $x = 2$. An additive integration constant would vanish under differentiation, so the nonzero residual $-\pi i/2$ is an erroneous branch term in the derivative, not an equivalent antiderivative representation.

Likely root cause. The diagnosis locates the failure in `sympy/integrals/meijerint.py`: integration by parts delegates $\int \log(x - 1)/x dx$ to Meijer integration, whose shifted-log rewrite selects a branch that differentiates to $\log(x - 1)/x + i\pi/x$. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-004-real-antiderivative-gains-an-imaginary-term/](#)